

Rule-Based or LLM Agent? Real-Time Behaviour Selection for an Emotion-Responsive Reachy Mini

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Abstract—Outline (written last):

- **Problem:** LLM agents promise flexible social behaviour but may be too slow for real-time emotional reactions.
- **Method:** rule-based controller vs. local LLM agent on the Reachy Mini, identical perception and actions, first in simulation, then on the robot.
- **Result and conclusion:** latency, resources and behavioural quality; answer to the research question.

I. INTRODUCTION

Social robots are expected to notice how a person feels and to react to it. Even a robot with a strongly reduced expressive channel can do so: observers rate the Reachy Mini as warmer and more sociable when it expresses positive affect through head and antenna motion alone [4].

Problem statement. The component that maps the perceived state of a person to the robot's reaction – the *behaviour-selection layer* – is usually implemented with hand-written rules. Rules are fast and predictable, but every situation has to be anticipated, and repeated situations lead to identical, mechanical reactions. Large language models (LLMs) used as tool-calling agents [6] offer an alternative: the reaction is chosen from a textual description of the situation and the interaction history, which promises more varied and context-sensitive behaviour. Frameworks such as LangChain [1] and local inference servers such as Ollama [2] make this possible without a cloud connection. However, LLM inference takes hundreds of milliseconds to seconds, requires considerable compute and memory, and does not guarantee valid or reproducible outputs. This conflicts with the timing requirement of social interaction: users prefer robot responses after about one second [5], and a reaction that comes too late is no longer read as a response. It is therefore unclear whether an LLM agent can replace rule-based behaviour selection on a small robot in real time, and what it offers in return for its cost.

Research question.

Can a local LLM agent select emotion-responsive behaviour on the Reachy Mini in real time, and what does it offer over a rule-based controller?

The first part is answered by the agent's reaction latency relative to the one-second threshold, the second by its resource demand and by the validity, consistency and appropriateness of the chosen reactions.

Contribution. We implement a behaviour architecture for the Reachy Mini [3] in which perception, action repertoire and motion execution are fixed and only the behaviour-selection layer is exchanged. We compare a rule-based controller with LLM-agent variants on identical, recorded input, first in simulation and then locally on the robot, and derive recommendations on when an agent is worth its cost.

II. STATE OF THE ART

A. Emotional Responsiveness in Social Robots

- Emotions expressed through head and antennas alone are perceived by observers [4].
- Response timing decisive; preference peaks at about 1 s [5].
- Derived requirements: contingency, appropriateness, non-mechanical variability.

B. Rule-Based Behaviour Control

- State machines, behaviour trees, lookup tables.
- Fast, deterministic, verifiable – but rigid and repetitive.

C. LLM Agents for Robot Behaviour

- LLMs as planners and tool-calling agents [6]; local models via Ollama.
- Open issues: latency, invalid actions, non-reproducibility, resource demand.
- Research gap: no controlled comparison with rules for real-time emotional reactions on small hardware.

III. MATERIALS AND METHODS

A. Platform

- Reachy Mini: 6-DoF head, body rotation, two antennas, camera.
- Phase 1: simulation on a laptop; phase 2: Reachy Mini Wireless, everything running on the onboard computer (Raspberry Pi CM4, 4 GB RAM).

B. System Architecture

- Perception (MediaPipe): face position, distance, four emotion classes, hand gestures.
- Event filter: only relevant changes reach the behaviour selection.

- Motion layer (50 Hz): head keeps tracking the face; expressions are played without blocking.
- Shared action repertoire: seven emotional postures, wave, nod, no action.

C. Compared Configurations

- R: rule table (baseline).
- A-TC: LangChain agent with one tool per action.
- A-SO: single LLM call with schema-validated output.
- H: rules for time-critical events, LLM for the rest.
- Model: Qwen2.5 via Ollama; smaller variants on the robot.

D. Evaluation Design

- Input A: scripted event sequence (decision only); input B: recorded and annotated video (end to end).
- Ten interleaved runs per configuration, in simulation and on the robot.
- Metrics: decision and end-to-end latency (median, p95, share above 1 s); CPU/RAM; invalid actions; consistency across runs; appropriateness against annotation.

IV. EXPERIMENTAL RESULTS

Planned content – measurements pending.

- Latency per configuration, simulation vs. robot.
- Resource demand; does the agent fit on the robot next to perception?
- Validity, consistency and appropriateness of the reactions.
- Discussion: trade-off between latency/cost and flexibility; limitations.

V. SUMMARY AND OUTLOOK

- Answer to the research question and recommendation.
- Outlook: user study, latency reduction, memory across interactions.

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